SCHOOLCHILDREN'S MOVEMENT AND READINESS TO LEARN DURING EXTREME HEAT

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BACKGROUND

- Recess is the only opportunity during the school day for children to get a break, socialize, and move
- Extreme heat in Arizona (AZ) often prevents traditional outdoor recess
- No uniform guidance to support recess during extreme heat exists so school practices vary widely

METHODS

Setting: Phoenix Metro (AZ) elementary schools (N=5)

Design: Cross sectional observational

Participants: 5th grade/age 10-11 students (n=218) and staff (n=15)

Data: 15 interviews (Aim 1); 981 questionnaires over 6 weeks (Aim 2); 91 recess observations, 5 thermal assessments over 5 weeks (Aim 3)

Concurrent QUAL-quant mixed methods

Data Collection

Quantitative

- **Movement**: Systematic Observation
- **Readiness to learn:** Survey
- **Thermal Exposure:** MaRTy

Qualitative

- **Recess strategies**: Interviews
- **Cost of intervention:** Interviews

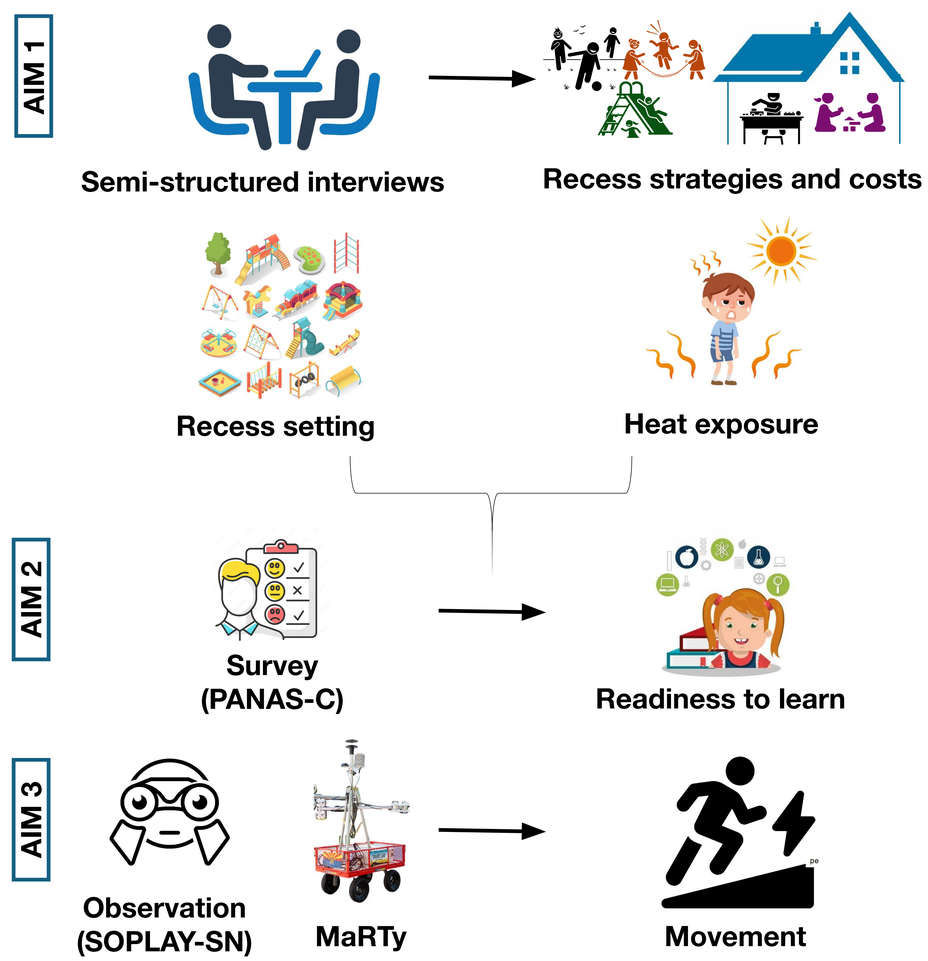
Data Analyses

- ANOVA; Cumulative link mixed models
- Matrix analysis

PURPOSE

To determine (1) common practices and costs of offering school recess during extreme heat, and differences in children's (2) readiness to learn and (3) movement by heat exposure and setting

APPROACH



RESULTS

- Air temperatures ranged from 91-109°F (33–43°C)
- Heat exposure, measured by mean radiant temperatures (air temperature, relative humidity, wind speed and direction), ranged from 103°F (39°C) under tree shade to 163°F (73°C) on courts
- Aim 1 - Essential resources to maintain safe outdoor recess (eg, shade structures, water bottle filling stations) required years of fundraising
- Aim 2 - Readiness to learn higher when recess was outside or in gyms compared to classrooms ($p<.05$)
- Aim 3 - Most recess was held outside (60%), with 20% in classrooms and 20% in gyms. Students were moderate-to-vigorously active 17% more outdoors and 40% more in gyms compared to recess inside classrooms ($p<.001$), where children were mainly sedentary (71%)

IMPLICATIONS

- Schools desire more schoolyard shade and dedicated indoor space for movement; Infrastructure improvements are expensive
- Kids are least ready for classroom instruction and mostly sitting when recess is inside
  - If recess were in gyms, kids would get ~5x more moderate-to-vigorous physical activity than classrooms and 2x more than outside
- The most active kids at recess maintain vigorous movement outside despite heat
- Outdoor spaces where kids are most active are generally least shaded

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Active Schools and Communities Lab